



MATH124: Big Data Homework 4

1) Conditioning a Gaussian

- distribution over random variables $x = (x_1, x_2)$ that is jointly Gaussian with parameters:

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix},$$

$$\text{where } \mu_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mu_2 = 5, \Sigma_{11} = \begin{bmatrix} 6 & 8 \\ 8 & 13 \end{bmatrix}, \Sigma_{21}^T = \Sigma_{12} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}, \Sigma_{22} = [14]$$

(a) compute marginal distribution $p(x_1)$

- the marginal distribution of x_1 asks: if we ignore x_2 what is the distribution of x_1 by itself?
- recall property of multivariate Gaussians:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \right)$$

- implies:

$$x_1 \sim \mathcal{N}(\mu_1, \Sigma_{11})$$

$$x_2 \sim \mathcal{N}(\mu_2, \Sigma_{22})$$

- so

$$p(x_1) = \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 & 8 \\ 8 & 13 \end{bmatrix} \right)$$

(b) compute marginal distribution $p(x_2)$

$$p(x_2) = \mathcal{N}(x_2 | \mu_2, \Sigma_{22}) = \mathcal{N}(x_2 | 5, 14)$$

(c) compute conditional distribution $p(x_1 | x_2)$

- jointly Gaussian variables:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \right)$$

- conditional distribution:

$$p(x_1 | x_2) = \mathcal{N}(x_1 | \mu_{1|2}, \Sigma_{1|2}), \text{ where } \mu_{1|2} = \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2)$$

- needed givens:

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$$

$$\mu_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \mu_2 = 5 \quad \Sigma_{11} = \begin{bmatrix} 6 & 8 \\ 8 & 13 \end{bmatrix}_{2 \times 2} \quad \Sigma_{12} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}_{2 \times 1} \quad \Sigma_{21} = \begin{bmatrix} 5 & 11 \end{bmatrix}_{1 \times 2} \quad \Sigma_{22} = 14_{1 \times 1} \quad \Sigma_{22}^{-1} = \frac{1}{14}$$

- calculate conditional mean

$$\mu_{1|2} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 5 \\ 11 \end{bmatrix} \left(\frac{1}{14} \right) (x_2 - 5)$$

$$\mu_{1|2} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 5/14 \\ 11/14 \end{bmatrix} (x_2 - 5)$$

$$\mu_{1|2} = \begin{bmatrix} \frac{5}{14}(x_2 - 5) \\ \frac{11}{14}(x_2 - 5) \end{bmatrix}$$

- calculate conditional covariance

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$$

$$= \begin{bmatrix} 6 & 8 \\ 8 & 13 \end{bmatrix} - \begin{bmatrix} 5 \\ 11 \end{bmatrix} \left(\frac{1}{14} \right) \begin{bmatrix} 5 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 8 \\ 8 & 13 \end{bmatrix} - \begin{bmatrix} 5/14 \\ 11/14 \end{bmatrix} \begin{bmatrix} 5 & 11 \end{bmatrix} \quad \begin{bmatrix} 5/14 \\ 11/14 \end{bmatrix} \begin{bmatrix} 5 & 11 \end{bmatrix} = \begin{bmatrix} 5/14(5) & 5/14(11) \\ 11/14(5) & 11/14(11) \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 8 \\ 8 & 13 \end{bmatrix} - \begin{bmatrix} 25/14 & 55/14 \\ 55/14 & 121/14 \end{bmatrix} \quad 6 - \frac{25}{14} = \frac{84}{14} - \frac{25}{14} = \frac{59}{14} \quad 8 - \frac{55}{14} = \frac{112}{14} - \frac{55}{14} = \frac{57}{14}$$

$$\Sigma_{1|2} = \begin{bmatrix} \frac{59}{14} & \frac{57}{14} \\ \frac{57}{14} & \frac{61}{14} \end{bmatrix}$$

$$\Sigma_{1|2} = \frac{1}{14} \begin{bmatrix} 59 & 57 \\ 57 & 61 \end{bmatrix}$$

$$p(x_1 | x_2) = \mathcal{N} \left(x_1 \left| \begin{bmatrix} 5/14(x_2 - 5) \\ 11/14(x_2 - 5) \end{bmatrix}, \frac{1}{14} \begin{bmatrix} 59 & 57 \\ 57 & 61 \end{bmatrix} \right. \right)$$

(d) compute conditional distribution $p(x_2 | x_1)$

- jointly Gaussian, conditional distribution:

$$p(x_2 | x_1) = \mathcal{N}(x_2 | \mu_{2|1}, \Sigma_{2|1})$$

- where:

$$\mu_{2|1} = \mu_2 + \Sigma_{21} \Sigma_{11}^{-1} (x_1 - \mu_1)$$

$$\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}$$

- givens: $\mu_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\mu_2 = 5$

$$\Sigma_{11} = \begin{bmatrix} 6 & 8 \\ 8 & 13 \end{bmatrix} \quad \Sigma_{12} = \begin{bmatrix} 5 \\ 11 \end{bmatrix} \quad \Sigma_{21} = \begin{bmatrix} 5 & 11 \end{bmatrix} \quad \Sigma_{22} = [14]$$

- calculate Σ_{11}^{-1}

$$\text{- recall } \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad \text{so,}$$

$$\Sigma_{11}^{-1} = \begin{bmatrix} 6 & 8 \\ 8 & 13 \end{bmatrix}^{-1} = \frac{1}{6(13) - 8(8)} \begin{bmatrix} 13 & -8 \\ -8 & 6 \end{bmatrix} = \frac{1}{43 - 64} \begin{bmatrix} 13 & -8 \\ -8 & 6 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 13 & -8 \\ -8 & 6 \end{bmatrix}$$

- calculate conditional mean

$$\mu_{2|1} = \mu_2 + \Sigma_{21} \Sigma_{11}^{-1} (x_1 - \mu_1)$$

$$\mu_{2|1} = 5 + \begin{bmatrix} 5 & 11 \end{bmatrix} \left(\frac{1}{14} \begin{bmatrix} 13 & -8 \\ -8 & 6 \end{bmatrix} \right) x_1$$

$$= \begin{bmatrix} 5 & 11 \end{bmatrix} \frac{1}{14} \begin{bmatrix} 13 & -8 \\ -8 & 6 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 5 & 11 \end{bmatrix} \begin{bmatrix} 13 & -8 \\ -8 & 6 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 5(13) + 11(-8) & 5(-8) + 11(6) \\ 5(-8) + 11(6) & 5(6) + 11(-8) \end{bmatrix} = \frac{1}{14} \begin{bmatrix} -23 & 26 \\ 26 & -23 \end{bmatrix}$$

$$= \frac{1}{14} \begin{bmatrix} -23 & 26 \\ 26 & -23 \end{bmatrix} = \begin{bmatrix} -\frac{23}{14} & \frac{26}{14} \end{bmatrix}$$

$$x_1 = \begin{bmatrix} x_{11} \\ x_{12} \end{bmatrix} \quad \text{so } \Sigma_{21} \Sigma_{11}^{-1} x_1 = \frac{1}{14} \begin{bmatrix} -23 & 26 \\ 26 & -23 \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{12} \end{bmatrix} = \frac{1}{14} (-23x_{11} + 26x_{12})$$

$$\text{conditional mean: } \mu_{2|1} = 5 + \frac{-23x_{11} + 26x_{12}}{14}$$

$$\mu_{2|1} = 5 - \frac{23}{14}x_{11} + \frac{26}{14}x_{12}$$

- calculate conditional variance

$$\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}$$

$$\Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12} = \frac{1}{14} \begin{bmatrix} -23 & 26 \\ 26 & -23 \end{bmatrix} \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$$= \frac{1}{14} \begin{bmatrix} -23(5) + 26(11) \\ 26(5) - 23(11) \end{bmatrix} = \frac{1}{14} \begin{bmatrix} -115 + 286 \\ 130 - 253 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 171 \\ -123 \end{bmatrix}$$

$$\Sigma_{2|1} = 14 - \frac{171}{14}$$

$$\Sigma_{2|1} = \frac{196}{14} - \frac{171}{14}$$

$$\Sigma_{2|1} = \frac{25}{14}$$

- therefore,

$$p(x_2 | x_1) = \mathcal{N} \left(x_2 \left| 5 - \frac{23}{14}x_{11} + \frac{26}{14}x_{12}, \frac{25}{14} \right. \right)$$

2) MNIST images: $28 \times 28 \Rightarrow 784$ pixel features

GOAL: predict handwritten digit

(a) restricts digits to 0 and 1 $\Rightarrow$ binary classification $\Rightarrow$ binary logistic regression